Prob. 5. (2 ST. Ch. 7, 2.3)

Verify that each of the f-ns are harmonic in a domain $\Omega \subset \mathbb{R}^2$. Begin Ω, which is largest domain in $\mathbb{R}^2$ for which f-n is harmonic

(a) (i) $u(x,y) = 1 \in C^2(\mathbb{R}^2)$ & ✓

(ii) $\underbrace{u_{xx}}_{0} + \underbrace{u_{yy}}_{0} = 0 + 0 \equiv 0$ ✓

Both prop-s che ck, so $\Omega = \mathbb{R}^2$ for $u = 1$.

(b) (i) $u(r,\theta) = r^n \cos(n\theta) \in C^\infty(\mathbb{R}^2)$

Note

$$\frac{\partial}{\partial r}\left[r \frac{\partial u}{\partial r}\right] = n^2 r^{n-1} \cos(n\theta) \quad \&$$

$$\frac{\partial^2 u}{\partial \theta^2} = -n^2 r^n \cos(n\theta)$$, So, the eq-n

$$\Delta u = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \cdot \frac{\partial^2 u}{\partial \theta^2} =$$

$$= \frac{1}{r} \cdot n^2 r^{n-1} \cos(n\theta) + \frac{1}{r^2} \cdot (-n^2) \cdot r^{n-2} \cos(n\theta) =$$

$= 0 + 0 \equiv 0$ ✓ Hence harmonic &

$\Omega = \mathbb{R}^2$.

(c) $u = r^n \sin(n\theta) = \text{pol-y} * \sin(\cdot) = \text{cont-s}$

(i) $u = r^n \sin(n\theta) \in C^\infty(\mathbb{R}^2)$ ✓

(ii) $\frac{\partial}{\partial r}\left(r \frac{\partial u}{\partial r}\right) = n^2 r^{n-1} \sin(n\theta)$

$\frac{\partial^2 u}{\partial \theta^2} = -n^2 r^n \sin(n\theta)$, then

$\Delta u = \frac{1}{r} \cdot n^2 \cdot r^{n-1} \sin(n\theta) + \frac{1}{r^2} \cdot (-n^2 \cdot r^n \sin(n\theta))$

$= 0$ ✓

(d) $u = \ln(r)$

(i) $u = \ln(r) \in C^\infty(\mathbb{R}^2 \setminus \{(0,0)\})$

(ii)

$u_r = \frac{1}{r}$ & $\frac{\partial}{\partial r}\left(r \frac{\partial u}{\partial r}\right) = 0$, $\frac{\partial^2 u}{\partial \theta^2} = 0$

$\Delta u = \frac{1}{r} \cdot n^2 \cdot (0) + 0 = 0$ ✓

hence $\Omega = \mathbb{R}^2 \setminus \{(0,0)\}$

(e)

(i) $u = r^{-n} \cos(n\theta) = \frac{1}{r^n} \cos(n\theta) \in C^\infty(\mathbb{R}^2 \setminus \{\underline{0}\})$

(ii) $\frac{\partial}{\partial r}\left[r \frac{\partial u}{\partial r}\right] = n^2 r^{-n-1} \cos(n\theta)$, $\frac{\partial^2 u}{\partial \theta^2} = -n^2 r^{-n} \cos(n\theta)$

$$\Delta u = \frac{1}{r}\frac{\partial}{\partial r}\left[r\frac{\partial u}{\partial r}\right] + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} =$$

$$= \frac{1}{r}\cdot n^2 r^{-n-1}\cdot\cos(n\theta) + \frac{1}{r^2}\cdot\left(-n^2 r^{-n}\cos(n\theta)\right)$$

$= A - A = 0$ ✓, hence u is harmonic on $\Omega = \mathbb{R}^2\setminus\{\underline{0}\}$.

f) (i) $u = r^{-n}\sin(n\theta) = \frac{1}{r^n}\sin(n\theta) \in C^\infty(\mathbb{R}^2\setminus\{\underline{0}\})$ &

(ii) $\frac{\partial}{\partial r}\left[r\frac{\partial u}{\partial r}\right] = n^2 r^{-n-1}\sin(n\theta)$ &

$\frac{\partial^2 u}{\partial \theta^2} = -n^2 r^{-n}\sin(n\theta)$ so

$$\Delta u = \frac{1}{r}n^2 r^{-n-1}\sin(n\theta) + \frac{1}{r^2}\left(-n^2 r^{-n}\sin(n\theta)\right) =$$

$= B - B = 0$ ✓, thus u is harm. on $\Omega = \mathbb{R}^2\setminus\{\underline{0}\}$. ∎